



ARTI 506 - Data Science and Analytics

Term 2 – 2025/2026

Project Proposal

Data-Driven Analysis of Tourism Demand and Spending Patterns in Saudi Arabia



AI Year 5

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Introduction

Tourism in Saudi Arabia has become an important contributor to economic growth and development. As part of the Saudi Vision 2030, the tourism sector aims to diversify the economy and promote regional growth. The Kingdom has set a new goal to attract 150 million tourists by 2030 [1]. Understanding tourism patterns is crucial for strategic planning, as tourist demand varies by visitor type (inbound or domestic), destination, and purpose of the visit. A key issue is that the high number of visitors to a particular region does not necessarily mean the highest tourism spending, because spending patterns are influenced by the purpose of the visit. This project aims to address this issue by analyzing tourism datasets of domestic and inbound tourists to uncover patterns in visitor distribution and spending behavior across different regions and visit purposes in Saudi Arabia from 2015 to the first half of 2025. This analysis will provide insights to support the development of data-driven tourism strategies aligned with Saudi Vision 2030.

Problem Statement

Tourism is a significant factor that affects the economy of kingdom of Saudi Arabia. Tourist demand varies by visitor type (inbound or domestic) and by destination and purpose of visit. The main problem is high number of visiting a certain region doesn't mean the highest spending rate because of the purpose of the visits. This gap shows the difficulty in deciding where to focus new investments or how to market other cities differently to increase the tourism movement in Saudi Arabia. This project aims to analyze tourism datasets for both inbound and domestic to show patterns in visitor distribution and spending behavior from 2015-2025H1.

Objectives

This project aims to explore, merge, and analyze four datasets related to the tourism sector in Saudi Arabia. It seeks to provide a comparison between inbound and domestic tourists, generate insights into tourism trends, and forecast future tourism patterns. To predict future tourism, machine learning algorithms will be applied, while exploratory data analysis (EDA) will be conducted to identify trends and patterns within the data. Additionally, data visualization techniques will be utilized to facilitate user interaction with the EDA findings and the predictions generated by the machine learning models.



Dataset description & Justification

For our project, we are using four datasets from the Saudi Open Data Portal that track the development of the Kingdom tourism sector from 2015 to 2025 H1. These datasets include detailed records of inbound and domestic tourists by their main purpose (such as religious, leisure, or business) [2][3], as well as spending and arrival statistics broken down by specific provinces and regions [4][5]. We are using this data to study how tourism trends are changing. By looking at both visitors number and spending, we can see which trip purposes, like religious or leisure travel, contribute most to the economy and how those patterns differ between local and international tourists.

This project aligns with Saudi Vision 2030 which aims for 150 million visitors and a 10% GDP contribution from tourism by the end of the decade. Using ten years of data, we can apply predictive modeling to forecast demand and identify emerging destinations, providing actionable insights for sustainable planning.

Initial analysis of the dataset's potential

After going through the four tourism datasets from the Saudi Open Data Platform, we believe they have strong potential for our project. The data covers a long period from 2015 to the first half of 2025 and includes important details like visitor type (inbound and domestic), main purpose of visit, region (13 administrative regions), number of tourists, and their spending. This helps us look at tourism from more than one side, not just how many people visit, but also how much they actually spend, which is the main issue we are trying to study. Since the data is organized by year and region, we can easily merge the datasets and calculate useful indicators such as spending per tourist and growth over time. Because the data spans almost ten years, it also allows us to apply forecasting models to predict future tourism trends. Even though the data is yearly and doesn't show detailed seasonal patterns, it is official, reliable, and detailed enough to support strong exploratory analysis and machine learning models that align with Vision 2030 goals.



References

- [1] “Saudi Vision 2030: Transforming Tourism for the Future.”
<https://www.visitsaudi.com/en/stories/vision-2030>
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<https://open.data.gov.sa/en/datasets/view/fbe7e81f-8c68-4fa7-89ff-ea3ed7bf3d5d> (accessed Feb. 15, 2026).
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